

Calculus 2 - Homework 6 (Problems)

1.

Classify $\sum_{n=1}^{\infty} (-1)^{n+1}/(2n - 1)$ as absolutely convergent, conditionally convergent, or divergent.

2.

Evaluate $\int \sec^3(x) \, dx$.

3.

Use the Midpoint Rule with $n = 4$ subintervals to approximate $\int_0^2 x^2 \, dx$. Report the estimate.

4.

Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} 2^n/n!$ converges or diverges.

5.

Evaluate $\int (\ln x)/x \, dx$.

6.

Determine whether $\int_1^{\infty} 1/x^2 \, dx$ converges, and if so find its value.

7.

Evaluate $\int x \cos(x) \, dx$.

8.

Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} n/2^n$ converges or diverges.

9.

Classify $\sum_{n=1}^{\infty} (-1)^{n+1}/n$ as absolutely convergent, conditionally convergent, or divergent.

10.

Evaluate $\int \ln(x) \, dx$.